

96%

48/50

OBSERVATION CHECKLIST

ABERDARE TECHNICAL COLLEGE

MODULE 4: DATABASE DESIGN — PRACTICAL 1

INFORMATION COMMUNICATION TECHNOLOGY

ELEMENT COVERED: ELEMENT 2 : CREATE COMPUTERIZED DATABASE

Duration: 3 Hours

Total Marks: 50

CANDIDATE AND ASSESSOR DETAILS

Trainee Name: KAIRH MARCUS Admission No.: 878

Date: 28/9/2026 Assessor Name: JAMES

Venue: LAB

CUTTING LIST (Tools/Materials Required)

Item	Quantity	Provided By
Computer with MS Access installed	1	Institution
ERD drawing tool (draw.io/Lucidchart) or plain paper & pen	1	Trainee/Institution
Printer and printing paper (for printed ERD, if applicable)	1	Institution
Flash disk / external storage	1	Trainee
Pen and pencil	1	Trainee

TASK 1: DRAW AN ENTITY RELATIONSHIP DIAGRAM (ERD) — 20 Marks

Item to be Evaluated	Allocated Marks	Candidate Score	Remarks
STUDENT entity correctly named and drawn	1	1	Entity named and drawn
COURSE entity correctly named and drawn	1	1	
All 4 relevant attributes correctly listed for STUDENT	3	2	7th attributes
All 4 relevant attributes correctly listed for COURSE	3	2	2 attributes listed
Primary Key correctly underlined for STUDENT	1	1	PK underlined in student
Primary Key correctly underlined for COURSE	1	1	underlined indicated
Relationship line correctly connects the two entities and is properly labeled	2	2	Entities well connected
Direction of relationship matches the scenario	1	1	Relationship matches the scenario

Cardinality "1" correctly placed on the STUDENT side	1	07	
Cardinality "M" correctly placed on the COURSE side	1	07	
Standard ERD symbols used consistently for entities, attributes, and relationship line	5	05	
Task 1 Total	20	18	

TASK 2: CREATE TABLES — 15 Marks

Item to be Evaluated	Allocated Marks	Candidate Score	Remarks
STUDENT table created in MS Access with field names matching the ERD	2	02	table Student created
COURSE table created in MS Access with field names matching the ERD	2	02	
Primary Key correctly set on STUDENT table	1	01	PK done &

Primary Key correctly set on COURSE table	1	01	FK set correctly in COURSE
Correct data type and appropriate field size assigned to every column	3	03	correct data type
FK field added to STUDENT table and correctly linked to COURSE table's PK <i>course</i> <i>students</i>	3	03	
FK data type matches the related PK data type	1	01	
At least 3 sample records inserted into STUDENT table	1	01	3 sample records added
At least 3 sample records inserted into COURSE table	1	01	
Task 2 Total	15	15	

TASK 3: CREATE A ONE-TO-MANY RELATIONSHIP — 15 Marks

Item to be Evaluated	Allocated Marks	Candidate Score	Remarks

Relationship correctly created in the Relationships window linking STUDENT and COURSE with correct field pairing (PK to FK)			
	03	03	Relationship correctly created
One-to-many symbol correctly displayed	1	01	1 to m symbol displayed
"Enforce Referential Integrity" option ticked	2	02	Referential integrity enforced
Relationship saved without errors	1	01	Relationship saved
Query created with correct fields selected from both tables	2	02	
Correct join type used, with results accurately showing each course and its registered students	3	03	
Clear, legible screenshots taken of the relationship window and query results	2	02	
Screenshots saved within the submitted file	1	01	Screenshots saved
Task 3 Total	15	15	

TOTAL PRACTICAL SCORE (Tasks 1–3): 48 / 50

PRODUCT CHECKLIST — 5 Marks

Item to be Evaluated	Allocated Marks	Candidate Score	Remarks
Filename matches Admission Number exactly	1	07	
All three tasks present and fully attempted in the submitted file	2	07	
Work is neatly organized, with clear and legible screenshots/evidence	1	07	
File submitted within the 3-hour deadline	1	00 07	
Product Checklist Total	5	04	

COMPETENCY DECISION

Total Practical Score: 48 / 50

☒ **COMPETENT** — Above 50

☐ **NOT YET COMPETENT** — Score of 49 or below

ASSESSOR'S COMMENTS

task completed successfully

Assessor's Name: James Signature: [Signature] Date: 28/09/2020

FEEDBACK FROM TRAINEE

Trainee's Signature: [Signature] Date: 28/09/2020

Task 1

2. Address Students and Courses

1. Student name & grade

Student ID

Course name & grade

Course ID



Student ID	Student name	Course name	Course ID
1	John	Mathematics	1
2	Jane	Mathematics	2
3	John	Mathematics	3

ABERDARE TECHNICAL COLLEGE
MODULE 4: DATABASE DESIGN

PRACTICAL 1

INFORMATION COMMUNICATION TECHNOLOGY
ELEMENT COVERED ELEMENT 2: CREATE COMPUTERIZED DATABASE

Duration: 3 HOURS

GENERAL INSTRUCTIONS

- ✓ *Read the scenario below carefully before attempting the task.*
- ✓ *Answer all three (3) tasks. Each task is compulsory.*
- ✓ *You may use pen and paper, a whiteboard tool, draw.io, Lucidchart, or MySQL Workbench for the ERD*
- ✓ *Use MS Access for database creation*
- ✓ *Label all entities, attributes, primary keys (PK), and foreign keys (FK) clearly on your ERD.*
- ✓ *Save your work with your Admission Number as the file name and submit as instructed*
- ✓ *If you chose to DRAW ERD using any software like draw.io, print and present the diagram to the assessor*

SCENARIO

Aberdare Technical College wants to computerize its student course registration system. Each STUDENT registers for one ~~or more~~ COURSE. Each course can have many students registered (a one-to-many relationship from STUDENT to COURSE).

TASK 1: DRAW AN ENTITY RELATIONSHIP DIAGRAM (ERD)

- Identify two (2) entities from the scenario
- List at least four (4) relevant attributes for each entity.
- Underline the Primary Key (PK) for each entity.
- Draw the relationship line between the two entities and label it with the relationship name
- Indicate the cardinality/multiplicity clearly using correct ERD notation (1 and M, or crow's foot notation) to show the one-to-many relationship.
- Use standard ERD symbols.

TASK 2: CREATE TABLES

- Based on your ERD, create the STUDENT table and the COURSE table in MS Access ✓
- Each table must have a clearly defined Primary Key. ✓
- Assign an appropriate data type and size to every column ✓
- The many side entity should have the foreign key for creating relationship. ✓
- Insert at least three (3) sample records into each table. ✓

TASK 3: CREATE A ONE-TO-MANY RELATIONSHIP

- Establish a working one-to-many relationship between STUDENT and COURSE using the Foreign Key created in Task 2.

- Enforce referential integrity
- Demonstrate that the relationship works by running a query (or showing a relationships diagram) that lists each course together with all students registered under it.
- Show or export evidence of the relationship by taking a screenshot and save it in the file you created.

END OF TASK